



CIQS - Analytical Quantum Circuit Compilation

The 'million-qubit compiler' by **arcadialab.fr**

- Extra quick tech brief -

For years, IBM defined the reference for quantum compilation and has recently published Benchpress, an open source tool that allows anybody to run the same benchmark suite their 2025 paper covers [1].

Among the 5 SDKs tested on the transpilation category, only Qiskit, the IBM flagship, passed all of their 892 tests, and did it in **17 hours** and 15 minutes [2]. All other SDKs failed some of the tests, and increased exponentially the completion time.

The exact transpilation suite was run independently with **CIQS**, on older, less powerful hardware. **CIQS** passed all 892 tests, with zero failures, in **75 minutes** [3].

Table 1 - CIQS ran on AMD Ryzen 5, 32 GB RAM [3]
All other SDKs ran on AMD Ryzen 9, 128 GB RAM [1]

SDK	Attempted	Passed	Failed	Total wall time no time out
CIQS	892	892	0	1h 15m
Qiskit	892	892	0	17h 15m
QTS	892	877	15	at least 35h 18m
Tket	892	814	78	at least 33h 11m
BQSKit	892	716	176	at least 64h 08m
Staq*	492	490	2	at least 13h 18m

*Staq covers half the test set by intentional scope (QASMBench only, no HamLib).

Per-test runtime geometric mean: CIQS at 0.051s, Qiskit at 0.310s
In the 200 to 433 qubit range, **CIQS** is 14.7× faster than Qiskit [2].

Architecture [4]

CIQS is a fully analytic compilation pipeline composed of:

- **CIQM** (mapper/router)
- **CIQO** (optimizer).

Every step (placement, routing, optimization) is determined by an exact criterion derived from the structure of the input circuit. There are no heuristics, no stochastic components, no tunable parameters, and no objective functions being minimized. **The same input produces the same output on every run.**

The pipeline is hardware-agnostic and handles quDits of any dimension natively.

*IBM quantum hardware was chosen by default as their platform was the only one to offer public, free QPU access when **CIQS** hardware validation was made.*

Scaling [4]

The IBM Benchpress abstract transpilation suite covers circuits up to 930 qubits. **Table 2** below extends from there using the same Benchpress harness and circuit type (Heisenberg model), on the same 2019 hardware.

All circuits were verified valid by the harness.

Qubits	Topology	2Q Gates	SWAPs	Gates Removed	Runtime
1,000	linear	4,995	0	2,997	0,10s
1,000	heavy-hex	7,582	2,587	2,997	0,13s
10,000	linear	49,995	0	29,997	1,2s
10,000	heavy-hex	72,792	22,797	29,997	1,7s
100,000	linear	499,995	0	299,997	9,6s
100,000	heavy-hex	739,658	239,663	299,997	32,5s
1,000,000	linear	4,999,995	0	2,999,997	93,4s
1,000,000	heavy-hex	7,513,506	2,513,511	2,999,997	848,8s

On the linear topology, runtime scales proportionally with qubit count across three orders of magnitude: 10× the qubits, 10× the time.

Hardware Validation [5]

The Hayden-Preskill circuit models black hole evaporation as a quantum system across multiple entangled registers over several steps. No experiment ever successfully ran this circuit, until now.

Paired with **CIQA** [6] (*see the CIQA tech brief*), **CIQS** compiled and routed the circuit onto IBM Heron r2, a heavy-hex topology that never was a design target for it.

The routing required 987 SWAP insertions and 120 gate removals across 300 circuits per run. The experiment ran across three IBM backends at two system sizes, with up to 307,200 shots per job. The mutual information $I_2(D:R)$ remained within ± 0.004 of zero across all runs and all time steps, indicating **no measurable noise introduced by the compilation**. The results have been reproduced a dozen times.

Full dataset, including IBM QPU side, published alongside the paper.

Links

- Compiled source code (free for non-commercial use):
<https://github.com/GamesOfArcadia/ciqs>
 - Presentation paper (architecture, correctness proofs, benchmarks):
<https://zenodo.org/records/19056796>
 - Benchpress comparison (full 1:1 results against the IBM SDK suite):
<https://zenodo.org/records/19896662>
 - Hayden-Preskill paper (circuit design, hardware runs, full dataset):
<https://zenodo.org/records/20120426>
-

References

- [1] P. D. Nation et al., « Benchmarking the performance of quantum computing software, » Nature Computational Science 5, 427-435 (2025). arXiv:2409.08844.
- [2] IBM's Benchpress repository :
https://github.com/Qiskit/benchpress/tree/1.0/published_results/transpilation
- [3] J. Pensédent, « Analytic Quantum Compilation at Scale: CIQS vs Heuristic SDKs on IBM Benchpress ». arcadialab.fr. DOI: 10.5281/zenodo.1989666 (2026).
- [4] J. Pensédent, "CIQS: A Scalable Pipeline for Processing Quantum Circuits Beyond 1,000,000 Qubits," arcadialab.fr, DOI: 10.5281/zenodo.19056796 (2026).
- [5] J. Pensédent, "Page Curve Tested on Hardware: Entropy rises, Correlations don't," arcadialab.fr, DOI: 10.5281/zenodo.19473695 (2026).
- [6] J. Pensédent, « CIQA: Analytical 1:5 Quantum Error Correction, 200x Below Surface-Code Overhead" arcadialab.fr, DOI: 10.5281/zenodo.19405503 (2026).